



**Course Title:** AP Computer Science Principles

**Curriculum:** CodeHS - JavaScript

**Classroom:** LB6

**Teacher:** Ms. Ma. Elena Montes

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**Grade Level:** High School (AP)

**Course Length:** One Year 125 Hours

**School Year:** 2026-2027

### Course Description:

AP Computer Science Principles introduces students to the foundational concepts of computer science and explores the impact computing and technology have on society. The CodeHS course emphasizes creative problem solving and real-world applications. Students explore important topics in computing using their own ideas and creativity, create computational artifacts of personal value, and develop an interest in computer science that can support future study in the field.

### Prerequisites:

There are no official prerequisites required by CodeHS. However, ATS recommends that students complete MakerSpace before enrolling in AP Computer Science Principles. CodeHS also recommends taking Introduction to Computer Science prior to AP courses and completing a first-year high school algebra course. Students should be comfortable with functions, function notation, and the Cartesian coordinate system.

### Learning Environment:

The course uses a blended classroom approach with web-based and physical activities. Students write and run code in the browser, create websites and digital artifacts, and participate in collaborative exercises. Lessons include video tutorials, quizzes, example programs, programming exercises, free-response activities, collaborative creation projects, and research projects.

### Programming Environment and Resources:

Students use the CodeHS browser-based editor to write text-based and block-based JavaScript programs and use Processing.js for graphical programs. Access to a computer and high-speed Internet is required. An online textbook is available through CodeHS lesson plans

### Overarching Goals:

- Increase and diversify participation in computer science.
- Develop confidence using computer science as a tool to express ideas and solve problems.
- Understand core principles of computing and how computing continues to change the world.
- Develop computational artifacts to solve problems, communicate ideas, and express creativity.
- Collaborate with others to solve problems and develop computational artifacts.
- Explain the impact computing has on society, economy, and culture.
- Analyze artifacts, identify and correct errors, and explain how they function.
- Understand representation of data, abstractions in computation and modeling, and responsible use of technology.

## Course Objectives:

The CodeHS AP Computer Science Principles course is based directly on the College Board AP Computer Science Principles Framework.

### **Computational Thinking Practices:**

**Practice 1:** Computational Solution Design - Design and evaluate computational solutions for a purpose.

**Practice 2:** Algorithms and Program Development - Develop and implement algorithms.

**Practice 3:** Abstraction in Program Development - Develop programs that incorporate abstractions.

**Practice 4:** Code Analysis - Evaluate and test algorithms and programs.

**Practice 5:** Computing Innovations - Investigate computing innovations.

**Practice 6:** Responsible Computing - Contribute to an inclusive, safe, collaborative, and ethical computing culture.

### **Five Big Ideas:**

- Big Idea 1: Creative Development
- Big Idea 2: Data
- Big Idea 3: Algorithms and Programming
- Big Idea 4: Computing Systems and Networks
- Big Idea 5: Impact of Computing

## Quizzes and AP Assessment:

At the end of most units, students take a summative multiple-choice unit quiz in the style of the AP Exam. The course also includes an AP Test Practice unit with a cumulative AP Practice Multiple Choice Test.

### **AP Exam Structure**

- **Section I:** 70 multiple-choice questions | 120 minutes | 70% of score.
- 57 single-select questions; 5 single-select questions with a reading passage about a computing innovation; 8 multiple-select questions where students select 2 answers.
- **Section II:** Create Performance Task written responses | 30% of score.
- Students submit program code, a video, and a student-authored Personalized Project Reference after 9 hours of in-class Create Performance Task time.
- Students answer 4 written-response prompts during 60 minutes of the end-of-course exam.

## AP Create Performance Task:

The Create Performance Task is an in-class assessment in which students design and implement a program to solve a problem, enable innovation, explore personal interest, or express creativity. The development process includes exploration, investigation, reflection, design, implementation, and testing. The course provides practice performance tasks and sets aside sufficient time for preparation and completion.

## Course Breakdown:

**Unit 1:** Introduction to Programming with Karel the Dog

**Unit 2:** Practice PT: Pair-Programming Paint

**Unit 3:** Programming with JavaScript

**Unit 4:** JavaScript Control Structures

**Unit 5:** Functions and Parameters

**Unit 6:** Practice PT: Tell a Story

**Unit 7:** Basic Data Structures

**Unit 8:** Digital Information

**Unit 9:** Practice PT: Steganography

**Unit 10:** Practice PT: Create Your Own Image Filter

**Unit 11:** The Internet

**Unit 12:** Practice PT: The Effects of the Internet

**Unit 13:** Data

**Unit 14:** Practice PT: Present a Data-Driven Insight

**Unit 15 & 16:** Explore MCQ Practice and Create Performance Task

**Unit 17:** Review for the AP Exam

**Unit 18:** Creative Development